

# Sparse-Form Self-Improvement at the Endpoint $s = \infty$

Automated mathematical discovery packet for arXiv:2003.02233

model: GPT5.6; lane: agent\_lane\_03

22 July 2026

## Abstract

Lorist and Nieraeth conjectured that sparse form domination with outer exponent  $q_0$  self-improves to every  $0 < q \leq q_0$ . They stated that even the case  $m = 1$ ,  $r = 1$ ,  $q_0 = 1$ ,  $s = \infty$  was unknown. We prove the full  $s = \infty$  slice, for arbitrary arity, arbitrary positive input averaging exponents, and arbitrary  $q_0 > 0$ . The proof splits  $|T(\vec{f})|$  at a multiple of the multilinear maximal function. The near part is pointwise maximal. On the far part, a nonlinear test function converts the assumed  $q_0$ -form into the desired  $q$ -form; the negative exponent  $q - q_0$  supplies the decisive inequality. The finite- $s$  part of the conjecture remains open.

## 1 Source conjecture and the solved slice

For a cube  $Q \subset \mathbb{R}^d$ , write

$$A_Q(\vec{f}) := \prod_{j=1}^m \langle |f_j| \rangle_{r_j, Q}.$$

A collection  $\mathcal{S}$  is sparse if there are pairwise disjoint measurable  $E_Q \subset Q$  with  $|E_Q| \geq |Q|/2$ . Conjecture 6.2 of Lorist–Nieraeth [1] asks whether sparse form domination at  $q_0$  and  $s \in (q_0, \infty]$  implies the analogous domination at every smaller  $q$ . Using any other fixed sparseness parameter only changes constants: an  $\eta$ -sparse family can be split into finitely many half-sparse families, with the number depending only on  $\eta$ . The source conjecture and its explicitly simplest unknown case appear on page 25 of the arXiv PDF:

However, at this point it is not clear how to deal with the cases  $t \in (r, q_0]$ . In the case of (6.1) we can simply apply the estimate  $\|\cdot\|_{\ell^{q_0}} \leq \|\cdot\|_{\ell^t}$  to obtain the domination required to complete the argument. However, if we only assume the sparse domination in form (6.3), it is unknown whether we automatically also have (6.3) with  $q_0$  replaced by a smaller exponent  $0 < q \leq q_0$ , meaning that it is not clear whether we have the flexibility to cover the cases  $t \in (r, q_0]$  or not without *assuming* that (6.3) also holds with  $q_0$  replaced by  $t$ .

We point out that replacing  $q_0$  by  $0 < q \leq q_0$  qualitatively yields the same weighted bounds, but the result is quantitatively worse in that it yields a worse exponent  $\gamma$  in the bound. Thus, on all accounts it seems that the following conjecture should hold:

**Conjecture 6.2** (Sparse form domination implies worse sparse form domination). *Let  $\vec{r} \in (0, \infty)^m$ ,  $q_0 \in (0, \infty)$ , and  $s \in (q_0, \infty]$ . Let  $T$  be an operator defined on  $m$ -tuples of functions and suppose that for any  $f_1, \dots, f_m \in L_c^\infty(\mathbf{R}^d)$  there exists a sparse collection  $\mathcal{S}$  such that*

$$\|T(\vec{f}) \cdot g\|_{L^{q_0}(\mathbf{R}^d)} \lesssim \left( \sum_{Q \in \mathcal{S}} \left( \prod_{j=1}^m \langle f_j \rangle_{r_j, Q} \right)^{q_0} \langle g \rangle_{\frac{1}{\frac{1}{q_0} - \frac{1}{s}}, Q}^{q_0} |Q| \right)^{\frac{1}{q_0}}.$$

*Then the same estimate also holds when we replace  $q_0$  by any  $q \in (0, q_0]$ .*

We point out that even the simplest case  $m = 1$ ,  $r = 1$ ,  $q_0 = 1$ ,  $s = \infty$  is unknown. For specific cases of  $T$  one can usually verify the conjecture by going back to the proof of (6.3)

We prove the conjecture for  $s = \infty$ . In this case the local exponent on the test function is  $q_0$  in the hypothesis and  $q$  in the conclusion.

For  $p > 0$ , define the supremal sparse form

$$\Phi_p(\vec{f}, g) := \sup_{\mathcal{S}} \left( \sum_{Q \in \mathcal{S}} A_Q(\vec{f})^p \int_Q |g|^p \right)^{1/p}, \quad (1)$$

where the supremum is over sparse collections. Passing between an existential sparse estimate and (1) loses at most an inessential factor two: if the supremum is finite and nonzero, choose a family attaining at least half of it. For the bounded, compactly supported data used below,  $\Phi_p$  is finite. This follows by separating cubes according to side length: sparseness bounds the total volume at bounded scales, while at large scales  $A_Q(\vec{f})^p \lesssim |Q|^{-p \sum_j 1/r_j}$  gives a convergent geometric tail. The zero case is immediate.

## 2 Proof intuition

Put  $F = |T(\vec{f})|$  and  $H = M_{\vec{r}}(\vec{f})$ . On  $\{F \leq C_T H\}$ , a standard stopping argument sparsely dominates  $H$  with the desired  $q$ -sum. On the complementary set, use the assumed  $q_0$  estimate with

$$h = F^{q/q_0 - 1} |g|^{q/q_0}.$$

The left side becomes the  $q$ -integral of  $Fg$ . On every selected cube,  $H \geq A_Q(\vec{f})$ ; hence the far-set inequality  $F > C_T H$  and  $q - q_0 \leq 0$  imply

$$A_Q(\vec{f})^{q_0} F^{q - q_0} \leq C_T^{q - q_0} A_Q(\vec{f})^q.$$

This is exactly the target sparse coefficient. Truncating  $h$  makes the use of the hypothesis rigorous.

### 3 A scalar maximal stopping lemma

**Lemma 1.** *For every  $q > 0$  and bounded compactly supported  $\vec{f}, g$ , one has*

$$\|M_{\vec{r}}(\vec{f})g\|_{L^q(\mathbb{R}^d)} \leq C_{d,m,\vec{r},q} \Phi_q(\vec{f}, g).$$

*Proof.* First restrict the supremum defining  $M_{\vec{r}}$  to a finite set of cubes in one dyadic grid. Start with its maximal cubes. For every selected cube  $Q$ , take as stopping children the maximal strict subcubes  $Q'$  for which, for some  $j$ ,

$$\langle |f_j| \rangle_{r_j, Q'} > (2m)^{1/r_j} \langle |f_j| \rangle_{r_j, Q}.$$

Assign each child to one such  $j$ . For a fixed  $j$ , disjointness and the definition of the average show that the assigned children have total measure at most  $|Q|/(2m)$ . Summing over  $j$  gives total child measure at most  $|Q|/2$ , so the stopping family is sparse.

Every cube whose stopping parent is  $Q$  has all its input averages controlled by the corresponding averages on  $Q$ . Therefore, pointwise,

$$M_{\vec{r}}^{\mathcal{D}}(\vec{f}) \leq C_{m,\vec{r}} \left( \sum_{Q \in \mathcal{S}} A_Q(\vec{f})^q \mathbf{1}_Q \right)^{1/q}.$$

After multiplication by  $|g|$ , raising to the  $q$ -th power and integrating, the right side becomes a sparse form bounded by  $\Phi_q(\vec{f}, g)^q$ . Increasing finite subcollections and monotone convergence handle the whole dyadic grid. Finally, the three-grid lemma dominates arbitrary cubes by one of finitely many dyadic grids; summing those finitely many estimates proves the claim.  $\square$

### 4 Endpoint self-improvement theorem

**Theorem 1.** *Let  $\vec{r} \in (0, \infty)^m$  and  $q_0 \in (0, \infty)$ . Suppose an operator  $T$  on  $m$ -tuples satisfies the following: for all bounded compactly supported  $\vec{f}, u$ , there is a sparse collection  $\mathcal{S}$  such that*

$$\|T(\vec{f})u\|_{L^{q_0}} \leq C_T \left( \sum_{Q \in \mathcal{S}} A_Q(\vec{f})^{q_0} \int_Q |u|^{q_0} \right)^{1/q_0}. \quad (1)$$

*Then, for every  $0 < q \leq q_0$  and all such  $\vec{f}, g$ , there is a sparse collection  $\mathcal{S}_q$  for which*

$$\|T(\vec{f})g\|_{L^q} \leq C_{d,m,\vec{r},q,q_0} C_T \left( \sum_{Q \in \mathcal{S}_q} A_Q(\vec{f})^q \int_Q |g|^q \right)^{1/q}.$$

*Thus Conjecture 6.2 holds in the full slice  $s = \infty$ .*

*Proof.* Enlarge  $C_T$  if necessary so that  $C_T \geq 1$ . Fix  $\vec{f}, g$ , and set  $F = |T(\vec{f})|$  and  $H = M_{\vec{r}}(\vec{f})$ . Split

$$E_0 = \{F \leq C_T H\}, \quad E_1 = \{F > C_T H\}.$$

By Lemma 1,

$$\int_{E_0} F^q |g|^q \leq C_T^q \|Hg\|_q^q \leq C_{d,m,\vec{r},q}^q C_T^q \Phi_q(\vec{f}, g)^q. \quad (2)$$

On  $E_1$ , define

$$h = F^{q/q_0-1} |g|^{q/q_0} \mathbf{1}_{E_1}, \quad h_N = \min\{N, h\}.$$

The negative power causes no problem:  $F > 0$  on  $E_1$ , and each  $h_N$  is bounded and supported inside the compact support of  $g$ . Apply (1) to  $h_N$  and then take the supremum over sparse families:

$$\begin{aligned} \int F^{q_0} h_N^{q_0} &\leq C_T^{q_0} \sup_{\mathcal{S}} \sum_{Q \in \mathcal{S}} A_Q(\vec{f})^{q_0} \int_Q h_N^{q_0} \\ &\leq C_T^{q_0} \sup_{\mathcal{S}} \sum_{Q \in \mathcal{S}} A_Q(\vec{f})^{q_0} \int_{Q \cap E_1} F^{q-q_0} |g|^q. \end{aligned}$$

For  $x \in Q \cap E_1$ , one has  $F(x) > C_T H(x) \geq C_T A_Q(\vec{f})$ . If the latter coefficient is positive, then  $q - q_0 \leq 0$  gives

$$A_Q(\vec{f})^{q_0} F(x)^{q-q_0} \leq C_T^{q-q_0} A_Q(\vec{f})^q;$$

if the coefficient is zero, its contribution is already zero. Consequently,

$$\int F^{q_0} h_N^{q_0} \leq C_T^q \Phi_q(\vec{f}, g)^q.$$

By monotone convergence,

$$\int_{E_1} F^q |g|^q \leq C_T^q \Phi_q(\vec{f}, g)^q. \quad (3)$$

Combining (2) and (3), and then selecting a sparse family whose  $q$ -form is within a factor two of the supremum in (1), proves the asserted existential sparse estimate. The dependence on  $C_T$  remains linear after taking the  $q$ -th root.  $\square$

## 5 Limitations, verification, and novelty

The finite- $s$  case is not claimed. Indeed, put

$$t_0 = (1/q_0 - 1/s)^{-1}, \quad t = (1/q - 1/s)^{-1}.$$

The same nonlinear test produces the local exponent  $u = qt_0/q_0$  on  $g$ . When  $s < \infty$  and  $q < q_0$ , one has  $u > t$ , so the source's desired local  $L^t$  average does not follow. This identifies the exact obstruction left by the proof.

The proof was audited for the cases  $q = q_0$ , zero sparse coefficients, negative powers, truncation, infinite suprema, and preservation of the existential sparse family. There is no computational dependency.

A bounded 22 July 2026 search covered the run's cheap indexes and deterministic source corpus, exact conjecture-title and number queries, the quoted simplest case, author/title variants, and arXiv searches involving smaller exponents and  $s = \infty$ . It found the source and a later thesis restating the conjecture as open, but no endpoint answer. Novelty confidence is moderate, subject to a full bibliographic review.

**Human-review recommendation.** Check Lemma 1, the admissibility and monotone limit of  $h_N$ , and the reversed inequality caused by  $q - q_0 \leq 0$ . These are the only nonformal steps.

## References

- [1] E. Lorist and Z. Nieraeth, *Sparse domination implies vector-valued sparse domination*, *Mathematische Zeitschrift* **301** (2022), 1–35; [arXiv:2003.02233](#).